

RL Components

Agent

- what?
 - Learner
 - Decision Maker
- Work
 - Observe
 - Think
 - Action
 - Learn
- Examples
 - Game Player
 - Robot
 - Self Driving Car
- Assignment
 - Move
 - Pickup
 - Carry
 - Deliver

Environment

- What?
 - World
 - Agent এর চারপাশ
- Work
 - Action গ্রহণ
 - State Update
 - Reward
 - Penalty
- Example
 - Game World
 - Room
 - Road
 - Grid World
- Assignment
 - Grid
 - ├─ Pickup
 - ├─ Goal
 - ├─ Agent Position
 - └─ Grid Size

State

- What?
 - Current Situation
 - Decision-এর ভিত্তি
- Contains
 - Agent Position
 - Pickup Position
 - Carrying Status
- Changes After
 - Move
 - Pickup
 - Delivery
- Different
 - Environment
 - Observation
- Assignment
 - Position
 - Pickup
 - Carrying

Action

- What?
 - Decision
 - Move
- Action Space
 - Allowed Actions
 - Fixed Set
- Types
 - Discrete
 - Continuous
- Assignment
 - Up
 - Down
 - Left
 - Right
 - Up-Left
 - Up-Right
 - Down-Left
 - Down-Right
- Result
 - New State

Episode

- What?
 - Start
 - Multiple Steps
 - Goal
 - End
- Training
 - Episode 1
 - Episode 2
 - Episode 3
 - Continue
- Assignment
 - Random Start
 - Pickup
 - Deliver
 - End
- Performance
 - < 20000 Episodes
 - ≤ 10 Steps

Policy

- What?
 - Decision Rule
 - State → Action
- Types
 - Random Policy
 - Optimal Policy
- Learns
 - Reward
 - Experience
 - Q-Table
 - Policy
- Assignment
 - Pickup
 - Goal
- Purpose
 - Best Action Select

Value & Q-Value

- Value (V)
 - State Quality
 - V(s)
- Q-Value (Q)
 - State + Action
 - Future Reward
 - Q(s,a)
- Agent
 - Compare Q
 - Choose Max Q
- Training
 - Reward
 - Update Q
 - Learn
- Assignment
 - Compare Actions
 - Best Move

Q-Table

- What?
 - Knowledge Base
 - Stores Q-Values
- Structure
 - Rows → States
 - Columns → Actions
 - Cells → Q-Values
- Training
 - Initial = 0
 - Reward
 - Update
 - Learn
- Decision
 - Compare Q
 - Choose Max Q
- Python
 - Q[state][action]